

Linus Dominic Nathaniel

Lead Data Engineer / Data Architect — Team Lead (4–6 Engineers)

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Location: Karachi, Pakistan

Status: Open to EU relocation — visa sponsorship required; also open to EU-remote

SUMMARY

Lead Data Engineer and Data Architect with 6+ years designing data platforms that turn fragmented operational systems into governed, queryable sources of truth. Currently **leading a team of 4–6 engineers** at Altair Capital Group, where I own the data architecture end-to-end — from source-system contracts and domain modeling through ingestion, transformation, BI and stakeholder enablement — most recently designing the firm's unified customer-journey model reconciling three source systems into a single funnel across ~2M entities. Strong across lakehouse design (Redshift, BigQuery, Athena on S3 parquet, Medallion modeling), the modern data stack (Apache Airflow, dbt, Spark, AWS Glue, Apache Kafka) and cross-system modeling with explicit lineage, contracts and governance. Product-company background (Pure Harvest Smart Farms, Arpatech) including **marketplace platform** data work on the Connection Cloud Marketplace.

LEADERSHIP

Team Lead — Altair Capital Group (Dec 2025 – Present)

Team size: **4–6 engineers** Scope: **Data platform, BI, integrations** Reporting: **Direct to leadership / CCO**

- Own technical direction for the data organisation: architecture review, sprint planning, code review standards, DAG/PR conventions, and the definition of done that the team builds against.
 - Spearheaded and delivered every major platform initiative below with the team — data lake re-architecture, unified customer-journey model, programmatic BI, AI conversation intelligence and the loan-pricing platform.
 - Mentored engineers into ownership of their own domains; contributed to hiring and technical screening.
 - Act as the translation layer between lending/operations stakeholders and engineering — gathering requirements, negotiating data contracts, and enabling self-serve analytics.
- **VentureDive** — Led the team-wide Talend → Apache Airflow platform migration and mentored junior engineers; codified Airflow, dbt, code-review and DAG-testing practice as team norms.
 - **Bleed AI** — Lead Support Engineer: led data-analysis effort across computer-vision delivery projects.

EXPERIENCE

Lead Data Engineer / Data Architect — Altair Capital Group

December 2025 – Present | Remote (US-based real-estate finance firm) | Team Lead — 4–6 engineers

- **Lead a team of 4–6 engineers** across data platform, BI and integrations — architecture ownership, sprint planning, code review, technical standards, mentorship and hiring input.
- Designed the firm's unified customer-journey domain model, reconciling three external source systems (parquet data lake, dialer logs and CRM) into a phone-keyed, six-stage funnel covering **~2M entities**; established it as the single source of truth for downstream BI and operations.
- Re-architected the production data lake on AWS S3 — canonical zoning, partitioning and cataloging across **~29 ingestion and transformation jobs**; eliminated long-standing duplication and set the governance pattern the team still follows.
- Defined and delivered the cross-system reconciliation framework between the loan-origination platform and the warehouse, surfacing data-quality gaps previously invisible to operations and enabling auditable diffs across millions of records.
- Established the firm's BI delivery model on Apache Superset — a version-controlled, programmatic deployment pipeline (Python plus Jinja-templated MySQL views) powering the Performance Score Ratings, Loan Officer pipeline and Daily Call dashboards used across sales, operations and leadership.
- Designed the voice and CRM integration pipeline on event-driven AWS Lambda (S3 pending-queue pattern, shared SSM-cached OAuth tokens), eliminating rate-limit failures at concurrent scale; wired into CloudWatch anomaly alerting on SNS.
- Architected the AI conversation-intelligence layer powering the firm's AI Telemarketing (AI-TM) agent: an LLM-driven Meaningful Conversation detector and role-aware call-note generator (Loan Officer, Processor, CCO) that writes structured insights back to CRM Leads and Contacts.
- Owned an internal loan-pricing platform end-to-end — from product requirements with the lending team through data modeling, calculation engine, CRM integration and term-sheet generation — for the firm's Bridge and DSCR loan products.

Senior Data Engineer — VentureDive

June 2025 – December 2025 | Karachi, Pakistan

- **Led** the modernization of the team's ETL platform from Talend to Apache Airflow, redefining standards for scheduling, observability and Git-based DAG versioning; reduced average pipeline runtime and on-call paging through parallelism redesign and SLA monitoring.
- Established the modern data stack reference architecture (Apache Airflow plus dbt plus warehouse) as the team-wide standard, with dbt models, tests, sources and lineage adopted across downstream transformations.
- Defined the multi-cloud deployment standard (Terraform plus GitHub Actions CI/CD) used across AWS, GCP and Azure client engagements (S3/Redshift, BigQuery/Dataflow, Synapse/Data Factory).
- **Mentored junior engineers** and codified Airflow patterns, code review and DAG-testing practices as team norms.

Data Engineer — Pure Harvest Smart Farms (product-based agri-tech company)

June 2023 – May 2025 | Remote

- Designed and deployed custom ETL pipelines on PySpark and AWS Glue for the company's own product platform, optimizing data workflows and reducing processing time ~30%.
- Automated multi-source ingestion into a centralized AWS data lake, increasing the analytics surface area for operations and growers.
- Implemented Medallion Architecture (bronze, silver, gold) on a Redshift-based lakehouse — data modeling, governance and performance tuning.
- Built interactive operational dashboards in Apache Superset for stakeholders across cultivation and supply chain.

Data and Machine Learning Engineer — Arpatech (Pvt) Ltd (product company — marketplace platform)

January 2022 – June 2023 | Karachi, Pakistan

- **Marketplace platform:** engineered the Custom Data Insights platform for the **Connection Cloud Marketplace**, a multi-vendor cloud-services marketplace — ingesting vendor and buyer transaction data and delivering interactive spend and comparative-analysis visualisations used by both sides of the marketplace to compare and price offerings.
- Built an Elasticsearch and Kibana querying app for the sales team, improving data retrieval speed ~50%.
- Developed and deployed a Smart Mirror IoT product for facial-recognition employee attendance, reducing manual tracking errors.

Lead Support Engineer and Data Analyst — Bleed AI

December 2019 – November 2021 | Karachi, Pakistan

- **Led** data analysis efforts for computer-vision projects, refining models for enhanced image-recognition accuracy.
- Optimized data pipelines, improving processing speed while maintaining high data integrity.

SELECTED PROJECTS

- **Unified Customer Journey Data Model** (Altair Capital Group) — Three external systems modeled onto a phone-keyed, six-stage funnel across ~2M entities; the conformed dimensions and bridge contracts powering all downstream BI.
- **Programmatic BI Platform on Apache Superset** (Altair Capital Group) — Version-controlled, code-deployed analytics (Python, Jinja-templated MySQL views, Superset API) powering the PSR, Loan Officer pipeline and Daily Call dashboards.
superset.altaircapitalgroup.com
- **Custom Data Insights — Connection Cloud Marketplace** (Arpatech) — Marketplace analytics platform delivering spend, vendor-comparison and category-level insight to buyers and vendors on a multi-vendor cloud-services marketplace.

SKILLS

- **Leadership and Delivery:** Team leadership (4–6 engineers), architecture review, code review standards, mentorship, sprint planning, stakeholder enablement, hiring and technical screening
- **Data Architecture and Governance:** Dimensional / Kimball modeling, data contracts, source-system contracts, lineage, MDM patterns, Medallion architecture, data domains, conformed dimensions · **AI and LLM:** LLM note generation, conversation intelligence, prompt engineering, role-aware agent design
- **Languages and Query:** Python, SQL, Bash · **Orchestration and Transformation:** Apache Airflow, dbt, Prefect, AWS Glue, AWS Step Functions
- **Processing:** Apache Spark, PySpark, Pandas, Apache Kafka, Hadoop · **Warehousing and Lakehouse:** Amazon Redshift, Google BigQuery, Snowflake, AWS Athena, Databricks
- **Cloud — AWS:** S3, Glue, Athena, Redshift, Lambda, RDS, Step Functions, EventBridge, ECS, Fargate, Secrets Manager · **GCP:** BigQuery, Dataflow, Cloud Storage · **Azure:** Data Factory, Synapse Analytics
- **Storage and Databases:** MySQL, PostgreSQL, MongoDB, Elasticsearch · **IaC and DevOps:** Terraform, Docker, GitHub Actions, CI/CD pipelines
- **BI and Visualization:** Apache Superset, Power BI, Grafana, Kibana, Streamlit · **CRM and Integrations:** Zoho CRM API, REST/JSON ingestion, Selenium scrapers

EDUCATION

Bachelor of Computer Science
Karachi Institute of Economics and Technology
August 2018 – August 2022

CERTIFICATIONS

- **Certified Data Engineer** — Karachi AI
- **Certified Data Analyst** — Karachi AI
- **Computer Vision and Image Processing** — Bleed AI

LANGUAGES

- English — Fluent
- Urdu — Native